INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

VOLTAGE STANDARD, DC

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for PD13WRGUEEA13 for all federal agencies.

1. SCOPE.

This commercial item description (CID) describes a DC Voltage Standard, hereinafter referred to as a Voltage Reference and *DC* Standard. The Voltage Reference and DC Standard provides accurate reference values of voltage for the calibration of test measurement and diagnostic equipment used in support and or development, repair and maintenance of navigational aids, radio frequency communications equipment and other measurement equipment. This CID lists the minimum performance, design, functional, environmental, reliability requirements and quality assurance provisions for the electronic calibrator.

2. CLASSIFICATION.

The Voltage Reference and DC Standard covered by this CID will be commercially available equipment that has factory installed options and/or may be modified to the extent necessary to meet the following description. The test set shall be Class 3, in accordance with MIL-PRF-28800, except as specified herein.

3. SALIENT CHARACTERISTICS.

3.1 Item Description. The Voltage Reference and DC Standard shall provide accurate direct current voltage reference values for the calibration, development, service & repair of test measurement and diagnostic equipment. This shall unit shall be configurable with four (4) independent “heads” with the capability of inter-comparison within the 4 head units and have

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external (rear) connectivity for service measurements and diagnostics. This instrument is typically used in rack configuration.

3.2 Voltage Reference and DC Standard. The Voltage Reference and DC Standard shall provide two static voltages 1.018 Volts and 10 Volts at current level of up to 12 mill ampere.

3.3 Stability Aging Accuracy. The Voltage Reference and DC Standard accuracy variance shall be ≤ ±2 ppm up to 10 Volts and ≤ ±1.0 ppm up to 1.018 Volts of programmed value within the temperature range of 32° to 122°F (0°C to 50°C) after tests in MIL-PRF-28800 section 4.5.5.1.1 has been performed for class 3 equipment classification (Per table 11 on page 41). The electronic calibrator stability shall be within the values listed in tables 1 thru 3 shown hereafter.

3.4 Temperature Coefficient. Voltage Reference and DC Standard shall have a temperature coefficient (TC) within the temperature range of 15 °C to 35 °C, the magnitude of TC is bounded by 10 Volts with a TC ≤0.04 ppm/°C and 1.018 Volts with a TC ≤0.1 ppm/°C.

3.5 Design and Construction. The Voltage Reference and DC Standard shall be used in both bench top and rack mounted environments. The Voltage Reference and DC Standard is a commercial off the shelf item subject to commercial products regulatory standards. The Voltage Reference and DC Standard shall comply with all applicable MIL-PRF-28800 sections. The Voltage Reference and DC Standard shall be designed and constructed in accordance with all applicable European Union (EU) requirements in order to have the “CE” marking affixed.

3.6 Display Indicators. The Voltage Reference and DC Standard shall have a method of indicating system status or conditions for power, calibration and charge or length of operation status.

3.7 System Safety. The Voltage Reference and DC Standard system shall comply with all sections of the Underwriters Laboratories (UL), IEC/ISO standard 61010 and 61010 for electrical safety.

3.8 Sound Levels. The Voltage Reference and DC Standard acoustic noise shall conform to sound level requirements on section 1 of ISO 7779 and section 2 of ISO 8253.

3.9 Electrostatic Discharge (ESD). The design of the Voltage Reference and DC Standard shall preclude equipment damage due to ESD (internal/external electrical arcing), protect personnel from electrical shock due to static charging, and prevent ignition of explosive atmospheres due to sparking in accordance with MIL-STD-1686 sections 5.2 and 5.7.2 and MIL-HDBK-263 for guidance.

3.10 Electro Magnetic Interference and Compatibility (EMI/EMC). The Voltage Reference and DC Standard EMI and EMC emissions shall be in accordance with the following radiated emission and susceptibility requirements of MIL-PRF-28800 section 4.5.6.5, MIL-STD-461: RE102 and RS103 & IEC-61000-4 thru IEC-61000-6. Emitted radiation absorption and radiation limits FCC Part 15.109 & 15.209.

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3.11 Port Damage. The Voltage Reference and DC Standard ports shall withstand shorted outputs indefinitely without damage to the instrument. The 10V output shall withstand voltages ≤ 220V dc, up to 25 mA continuous current for short periods of time ≤ 15 seconds.

3.12 Connectors. The Voltage Reference and DC Standard system shall have a "Banana" type connector set for two voltages with common ground and a chassis/ground terminal.

3.13 Operating Temperature Range. The Voltage Reference and DC Standard shall meet the specifications listed herein with full capabilities within the temperature range of 32° to 122°F (0°C to 50°C) after performing tests in MIL-PRF-28800 section 4.5.5.1.1 table 11 for class 3 equipment.

3.14 Storage Temperature Range. The Voltage Reference and DC Standard shall meet and/or exceed manufacturer's specifications for stored equipment after tests of MIL-PRF-28800 section 4.5.5.1.1 table 11 for temperature range and relative humidity storage for class 3 equipment classification.

3.15 Altitude Requirements. The Voltage Reference and DC Standard shall meet its operating performance and accuracy while operating at an altitude equal or greater than 6,000 ft. (1,828 meters). The equipment shall meet performance and accuracy requirements, for non-operating storage up to 10,000 (3,050 meters).

3.16 Calibration & Maintenance Adjustments. No special tools or individual instrument unique software shall be required for alignment or adjustment except those furnished with the electronic calibrator test set.

3.17 Calibration Period. The design of the Voltage Reference and DC Standard shall provide readily accessible calibration and maintenance adjustments. The calibration adjustments shall be accessible without removal of the instrument internal modules. The calibration interval shall be based on a period of one calendar year or an operating time of 8,766 hours or whichever occurs first.

3.18 Reliability. The design of the Voltage Reference and DC Standard shall be such that under normal use and operation the mean time between failures (MTBF) exceeds the minimum of 2,080 hours of operation accordance with MIL-PRF-28800. The manufacturer shall provide a summary of maintenance or warranty records to establish the validity of the statistical assertion in accordance with MIL-PRF-28800 section. The manufacturer shall be able to reasonably establish the claim that the offered equipment is capable of the required level of reliability by meeting and/or exceeding MIL-PRF-28800 sections and section.

3.19 Maintainability. The Voltage Reference and DC Standard shall meet the maintainability criterion in accordance with MIL-STD-1472 paragraphs 5.9 through 5.9.18 and 5.9.21 and MIL-PRF-28800 sections 3.14.1 through 3.14.4 and sections 6.6.2.4 and 6.6.24.1.

3.20 Preventive Maintenance. Preventive maintenance shall not require more than 15 minutes per 30-day period. Preventive maintenance shall not require breaking of the equipment seams where

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calibration seals would normally be placed in accordance with MIL-PRF-28800 sections 3.14.2 & 3.14.3.

3.21 Corrective Maintenance. The Voltage Reference and DC Standard shall be designed to meet reparability criterion in accordance with section 3.14.3 sub sections a through d of MIL-PRF-28800.

3.22 Electrical Power Sources & Connections. The Voltage Reference and DC Standard shall operate from nominal commercial, military, aircraft and power sources with a nominal voltage of 120 V_{RMS} Single Phase AC (±10%) at line frequencies of 50 Hz & 60 Hz (±5%), in accordance with MIL-PRF-28800 paragraph 3.1.5.2 and 3.1.5.2a. Transient state conditions shall meet MIL-PRF-28800 requirements for the classification specified herein. AC input power shall be used to power the SA and recharge batteries for nonvolatile memory if equipped with them.

3.23 Dimensions. The Voltage Reference and DC Standard shall comply with width and depth of EIA/ECA-310 standard equipment rack configuration. The width shall be the 19” when configured with rack adapter/mount hardware and include a rack-mount conversion kit if required at organizational level. The height shall be compatible with standard EIA height units. One unit or "U" is defined by EIA/ECA-310 to be 1.75 inches or 44.45 mm.

3.24 Weight. The Voltage Reference and DC Standard system shall comply with MIL-STD-1472 section 5.9.11.3.3 for weight & size restrictions for single person lift & handling. The total weight of the electronic calibrator, excluding accessories and manuals, shall not exceed a two-person lift of 70 pounds, (32 kilograms).

4. REGULATORY REQUIREMENTS.

4.1 Recycled recovered materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

4.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

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5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and the product conforms to the producer's engineering drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

5.2. Verification. Verification of the delivered electronic calibrator will be performed at the delivery point by the end user.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained online at https://assist.dla.mil/ or from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

7.1.2 ISO documents may be obtained online at http://www.iso.org/ or from American National Standards Institute, 1899 L St NW, 11th Floor US-Washington, DC 20036.

7.1.3 FAR documents may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of the FAR may be obtained online from https://www.acquisition.gov/far/ .

7.1.4 IEC documents may be obtained online at http://www.iec.ch/ or from IEC Regional Centre for North American (IEC-ReCNA) 446 Main Street, 16th Floor, Worcester, MA 01608.

7.1.5 EIA/ECA documents may be obtained online at http://www.global.ihs.com/ or from IHS Customer Care 15 Inverness Way East, Englewood CO. 80112.

7.16 FCC documents may be obtained online at http://www.ecfr.gov/ or from the Federal Communications Commission 445 12th Street SW, Washington DC 20554.

7.2 National Stock Number (NSN). The following NSN corresponds to this CID. The NSN may not be indicative of all possible NSNs associated with this CID. (For reference only)

NSN: 6625-01-393-4087

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7.3 Key words.

Accurate Reference
Direct Current, Stable
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Custodians:
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